

D - Chalkboard Median

分值: 400 分

Problem Statement

There is one integer X written on a blackboard.

黑板上初始写有一个整数 X 。

You are given Q queries to process in order. The i -th query ($1 \leq i \leq Q$) is as follows.

给定 Q 个询问，你需要按顺序处理它们。第 i 个询问 ($1 \leq i \leq Q$) 的内容如下。

Two integers A_i and B_i are given. Write two new integers A_i and B_i on the blackboard.

Then, output the median of the $2i + 1$ integers written on the blackboard.

给定两个整数 A_i 和 B_i ，在黑板上写下两个新整数 A_i 和 B_i 。

之后，输出当前黑板上所有 $2i + 1$ 个整数的中位数。

Constraints

- $1 \leq X \leq 10^9$
 - $1 \leq Q \leq 2 \times 10^5$
 - $1 \leq A_i, B_i \leq 10^9$
 - All input values are integers.
 - $1 \leq X \leq 10^9$
 - $1 \leq Q \leq 2 \times 10^5$
 - $1 \leq A_i, B_i \leq 10^9$
 - 所有输入值都是整数。
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Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

X
 Q
 $A_1 B_1$
 $A_2 B_2$
 $\vdots$
 $A_Q B_Q$

Output

Output Q lines.

输出 Q 行。

The i -th line should contain the answer to the i -th query.

第 i 行应输出第 i 个询问的答案。

Sample Input 1

```
5
3
2 3
1 2
8 9
```

Sample Output 1

```
3
2
3
```

In the first query, the integers written on the blackboard become 5, 2, 3, and their median is 3.

处理第一个询问后，黑板上的整数变为 5, 2, 3，它们的中位数是 3。

In the second query, the integers written on the blackboard become 5, 2, 3, 1, 2, and their median is 2.

处理第二个询问后，黑板上的整数变为 5, 2, 3, 1, 2，它们的中位数是 2。

In the third query, the integers written on the blackboard become 5, 2, 3, 1, 2, 8, 9, and their median is 3.

处理第三个询问后，黑板上的整数变为 5, 2, 3, 1, 2, 8, 9，它们的中位数是 3。

Sample Input 2

```
1
4
2 3
4 5
6 7
8 9
```

Sample Output 2

```
2
3
4
5
```

Sample Input 3

```
278117031
7
167642909 517897721
148434323 567739597
319926999 481642530
659199879 252516557
49913403 798318034
89701408 892537201
199166668 742285869
```

Sample Output 3

278117031

278117031

319926999

319926999

319926999

319926999

319926999